Discrete Structures / Mathematics

Lecture # 03 & 04

LAWS OF LOGIC

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Applying Laws of Logic

- Using laws of logic simplify the statement form.

$$p \vee [\sim(\sim p \wedge q)]$$

- **<u>Solution:</u>**

$\equiv p \vee [\sim(\sim p) \vee (\sim q)]$	DeMorgan's Law
$\equiv p \vee [p \vee (\sim q)]$	Double Negative Law
$\equiv [p \vee p] \vee (\sim q)$	Associative Law for $\vee$
$\equiv p \vee (\sim q)$	Indempotent Law

This is the simplified statement form.

EXAMPLE

- Using Laws of Logic, verify the logical equivalence.

$$\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$$

Solution:

$\equiv (\sim(\sim p) \vee \sim q) \wedge (p \vee q)$	DeMorgan's Law
$\equiv (p \vee \sim q) \wedge (p \vee q)$	Double Negative Law
$\equiv p \vee (\sim q \wedge q)$	Distributive Law in reverse
$\equiv p \vee c$	Negation Law
$\equiv p$	Identity Law

EXERCISES

- Show that $\sim (p \vee (\sim p \wedge q))$ and $\sim p \wedge \sim q$ are logically equivalent.

- Simplify $\sim (\sim (p \vee q) \wedge \sim q))$

SIMPLIFYING A STATEMENT:

- "You will get an A if you are hardworking and the sun shines, or you are hardworking and it rains."

- **Solution:**

  Let

  p = "You are hardworking'

  q = "The sun shines"

  r = "It rains"

- The condition is then $(p \wedge q) \vee (p \wedge r)$

$$(p \wedge q) \vee (p \wedge r)$$

- $\equiv p \wedge (q \vee r)$ Distributive law in reverse

- Putting $p \wedge (q \vee r)$ back into English, we can rephrase the given sentence as

- **"You will get an A if you are hardworking and the sun shines or it rains."**

EXERCISE:

- Use Logical Equivalence to rewrite each of the following sentences more simply.

- **1. It is not true that I am tired, and you are smart.**
  {I am not tired or you are not smart.}
- **2. It is not true that I am tired or you are smart.**
  {I am not tired and you are not smart.}
- **3. I forgot my pen or my bag and I forgot my pen or my glasses.**
  {I forgot my pen or I forgot my bag and glasses.
- **4. It is raining and I have forgotten my umbrella, or it is raining and I have forgotten my hat.**
  {It is raining and I have forgotten my umbrella or my hat.}

CONDITIONAL STATEMENT or IMPLICATION

- <u>Introduction</u>

  Consider the statement:

  "**If you earn an A in Math, then I'll buy you a computer.**"

- This statement is made up of two simpler statements:

  p: "You earn an A in Math," and

  q: "I will buy you a computer."

  if p is true, then q is true, or, more simply, **if** p, **then** q.

  We can also phrase this as p **implies** q, and we write **p** $\rightarrow$ **q**.

The original statement is then saying:

if p is true, then q is true

Or

more simply, **if** p, **then** q.

We can also phrase this as p **implies** q, and we write $\mathbf{p \rightarrow q}$.

TRUTH TABLE for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

CONDITIONAL STATEMENTS OR IMPLICATIONS:

- **Definition:**
- If p and q are statement variables, the conditional of q by p is "If p then q" or "p implies q" and is denoted $p \rightarrow q$.
- It is false when p is true and q is false; otherwise it is true.

- The arrow "$\rightarrow$ " is the **conditional** operator
- and in $p \rightarrow q$ the statement **p** is called the **hypothesis**
- **(or antecedent)**
- q is called the conclusion (or consequent).

PRACTICE WITH CONDITIONAL STATEMENTS:

- Determine the truth value of each of the following conditional statements:

"If 1 = 1, then 3 = 3"	TRUE
"If 1 = 1, then 2 = 3"	FALSE
"If 1 = 0, then 3 = 3"	TRUE
"If 1 = 2, then 2 = 3"	TRUE
"If 1 = 1, then 1 = 2 and 2 = 3"	FALSE
"If 1 = 3 or 1 = 2 then 3 = 3"	TRUE

ALTERNATIVE WAYS OF EXPRESSING IMPLICATIONS

- The implication **p** $\rightarrow$ **q** could be expressed in many alternative ways as:

"if p then q"	"not p unless q"
"p implies q"	"q follows from p"
"if p, q"	"q if p"
"p only if q"	"q whenever p"
"p is sufficient for q"	"q is necessary for p"

EXERCISE:

- Show that $(p \wedge q) \rightarrow (p \vee q)$ is tautlology

- Write the following statements in the form "**if p, then q**" in English.

  a) Your guarantee is good only if you bought your CD less than 90 days ago.

  If your guarantee is good, **then** you must have bought your CD less than 90 days ago.

  b) To get tenure as a professor, it is sufficient to be world-famous.

  If you are world-famous, **then** you will get tenure as a professor.

c) That you get the job implies that you have the best credentials.

If you get the job, **then** you have the best credentials.

d) It is necessary to walk 8 miles to get to the top of the Peak.

If you get to the top of the peak, **then** you must have walked 8 miles.

TRANSLATING ENGLISH SENTENCES TO SYMBOLS:

- **Let p and q be propositions:**

  p = “you get an A in the final exam”

  q = “you do every exercise in this book”

  r = “you get an A in this class”

- Write the following propositions using p, q, and r and logical connectives.

- To get an A in this class it is necessary for you to get an A in the final.

  SOLUTION $r \rightarrow p$

- You do every exercise in this book; You get an A in the final, implies, you get an A in the class.

  SOLUTION $p \wedge q \rightarrow r$

- Getting an A in the final and doing every exercise in this book is sufficient for getting an A in this class.

  SOLUTION $p \wedge q \rightarrow r$

TRANSLATING SYMBOLIC PROPOSITIONS TO ENGLISH

- Let **p**, **q**, and **r** be the propositions:

  p = “you have the flu”
  q = “you miss the final exam”
  r = “you pass the course”

- Express the following propositions as an English sentence.

- $p \rightarrow q$

  If you have flu, then you will miss the final exam.

- $\sim q \rightarrow r$

  If you don't miss the final exam, you will pass the course.

- $\sim p \wedge \sim q \rightarrow r$

  If you neither have flu nor miss the final exam, then you will pass the course.

HIERARCHY OF OPERATIONS FOR LOGICAL CONNECTIVES

$\sim$ (negation)

$\wedge$ (conjunction)

$\vee$ (disjunction)

$\rightarrow$ (conditional)

Construct a truth table for the statement form $(p \vee \sim q) \rightarrow \sim p$

p	q	~q	~p	$p \vee \sim q$	$(p \vee \sim q) \rightarrow \sim p$
T	T	F	F	T	F
T	F	T	F	T	F
F	T	F	T	F	T
F	F	T	T	T	T

Construct a truth table for the statement form $(p \rightarrow q) \wedge (\sim p \rightarrow r)$

p	q	r	$p \rightarrow q$	$\sim p$	$\sim p \rightarrow r$	$(p \rightarrow q) \wedge (\sim p \rightarrow r)$
T	T	T	T	F	T	T
T	T	F	T	F	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	F	T	T	F	F

LOGICAL EQUIVALENCE INVOLVING IMPLICATION

- Use truth table to show $p \rightarrow q \equiv \sim q \rightarrow \sim p$

p	q	~q	~p	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

same truth values

IMPLICATION LAW

- $p \rightarrow q \equiv \sim p \vee q$

p	q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

NEGATION OF A CONDITIONAL STATEMENT

- Since $p \rightarrow q \equiv \sim p \vee q$ therefore
- $\sim (p \rightarrow q) \equiv \sim (\sim p \vee q)$

  $\equiv \sim (\sim p) \wedge (\sim q)$ by De Morgan's law

  $\equiv p \wedge \sim q$ by the Double Negative law

- Thus the negation of "**if p then q**" is logically equivalent to "**p and not q**".

- **Note:**

  Accordingly, the negation of an if-then statement does not start with the word if.

EXAMPLES

- **Write negations of each of the following statements:**

- If Ali lives in Pakistan then he lives in Lahore.

  Ali lives in Pakistan and he does not live in Lahore.

- If my car is in the repair shop, then I cannot get to class.

  My car is in the repair shop and I can get to class.

- If x is prime then x is odd **or** x is 2.

  x is prime but x is not odd **and** x is not 2.

- If n is divisible by 6, then n is divisible by 2 **and** n is divisible by 3.

  n is divisible by 6 but n is not divisible by 2 **or** by 3.

INVERSE OF A CONDITIONAL STATEMENT

- The inverse of the conditional statement **p → q** is **~p → ~q**
- A conditional and its inverse are not equivalent as could be seen from the truth table.

p	q	p → q	~p	~q	~p →~q
T	T	T	F	F	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T

different truth values in rows 2 and 3

WRITING INVERSE

- ***If today is Friday, then 2 + 3 = 5.***

  If today is not Friday, then $2 + 3 \neq 5$.

- ***If it snows today, I will ski tomorrow.***

  If it does not snow today I will not ski tomorrow.

- ***If P is a square, then P is a rectangle.***

  If P is not a square then P is not a rectangle.

- ***If my car is in the repair shop, then I cannot get to class.***

  If my car is not in the repair shop, then I shall get to the class.

CONVERSE OF A CONDITIONAL STATEMENT

- The converse of the conditional statement $\mathbf{p} \rightarrow \mathbf{q}$ is $\mathbf{q} \rightarrow \mathbf{p}$
- A conditional and its converse are not equivalent.
- i.e., $\rightarrow$ is not a commutative operator.

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

not the same

WRITING CONVERSE

- *If today is Friday, then 2 + 3 = 5.*

  If 2 + 3 = 5, then today is Friday.

- *If it snows today, I will ski tomorrow.*

  I will ski tomorrow only if it snows today.

- *If P is a square, then P is a rectangle.*

  If P is a rectangle then P is a square.

- *If my car is in the repair shop, then I cannot get to class.*

  If I cannot get to the class, then my car is in the repair shop.

CONTRAPOSITIVE OF A CONDITIONAL STATEMENT

- To form the **contrapositive** of the **conditional statement**, interchange the hypothesis and the conclusion of the inverse **statement**.
- The **contrapositive** of "If it rains, then they cancel school" is "If they do not cancel school, then it does not rain.
- The contrapositive of the conditional statement $\mathbf{p} \rightarrow \mathbf{q}$ is $\sim \mathbf{q} \rightarrow \sim \mathbf{p}$
- A conditional and its contrapositive are equivalent. Symbolically $\mathbf{p} \rightarrow \mathbf{q} \equiv \sim\mathbf{q} \rightarrow \sim\mathbf{p}$

Example of Contrapositive

Statement	If it is raining, then the grass is wet	$p \rightarrow q$
Converse	If the grass is wet, then it is raining	$q \rightarrow p$
Inverse	If it is not raining, then the grass is not wet	$\sim p \rightarrow \sim q$
Contrapositive	If the grass is not wet, the it is not raining	$\sim q \rightarrow \sim p$

- *If today is Friday, then 2 + 3 = 5.*

  If $2 + 3 \neq 5$, then today is not Friday.

- *If it snows today, I will ski tomorrow.*

  I will not ski tomorrow only if it does not snow today.

- *If P is a square, then P is a rectangle.*

  If P is not a rectangle then P is not a square.

- *If my car is in the repair shop, then I cannot get to class.*

  If I get to the class, then my car is not in the repair shop.

INCLUSIVE AND EXCLUSIVE OR

- In English when we use OR in this sense, when we say p or q. It means either p is true, or q is true or possibly both are true.

- *Inclusive* OR allows both possibilities as well as either of them. So, if either A or B is True, or if both are True, then the statement value is True.

- Whereas *Exclusive* OR only allows one possibility. So if either A or B is true, then and only then is the value True. If both A and B are True, even then the statement's value will be False.

EXAMPLE

- **E.g.:** Give him a book or bag. It means either give him a book or bag or possibly give him both bag and book.
- In the above statement, we are using **OR in Inclusive** sense.

- **E.g.:** Tomorrow at 9’o clock, I will be in Lahore or Islamabad.
- In the above statement, we are using **OR in Exclusive** sense.

Truth table for EXCLUSIVE OR

$$(p \vee q) \wedge \sim (p \wedge q)$$

p	q	$p \vee q$	$p \wedge q$	$\sim (p \wedge q)$	$(p \vee q) \wedge \sim (p \wedge q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Exclusive OR

- When **OR** is used in exclusive sense, the statement "p or q" means "p or q but not both"

or

- "**p or q and not p and q**" which translates into symbols as:

$$(p \vee q) \wedge \sim (p \wedge q)$$

Symbols for EXCLUSIVE OR

$p \oplus q$

or

p XOR q

- **Note:**
  Basically
  $p \oplus q \equiv (p \vee q) \wedge \sim (p \wedge q)$

Examples of Implication

- **p** = “Ali lives in Lahore”
  q = “Ali lives in Pakistan”

  “if **p** then **q**”
  “**p** implies **q**”
  “if **p, q**”
  “**p** only if **q**”
  “**p** is sufficient for **q**”

Note:

- IMPLICATION is logically equivalent to its CONTROPOSITIVE.
- IMPLLICATION and its CONVERSE they are not equivalent to each other logically.
- Implication’s INVERSE and CONVERSE are logically equivalent to each other.
- IMPLICATION and its CONTROPOSITIVE are equivalent to each other.

Example of Contrapositive

Statement	If it is raining, then the grass is wet	$p \rightarrow q$
Converse	If the grass is wet, then it is raining	$q \rightarrow p$
Inverse	If it is not raining, then the grass is not wet	$\sim p \rightarrow \sim q$
Contrapositive	If the grass is not wet, then it is not raining	$\sim q \rightarrow \sim p$

Examples:

- If you work hard you get good grades.

  w = “You work hard”

  g = “You get good grades”

- $w \rightarrow g$ = If w then g

  = “If you work hard then you get good grades”.

  Converse: True

  $g \rightarrow w$ = If g then w

  = g only if w

  = “You get good grades only if you work hard.”

- If we sum up the implication ($w \rightarrow g$) and its converse ($g \rightarrow w$) we get

"You get good grades ***if and only if*** you work hard."

By summing up we get a new connective and we called this new connective **BICONDITIONAL**.

Bi-Conditional Connective

- If p and q are statement variables, the biconditional of p and q is

- "**p if, and only if, q**" and is denoted **p** $\leftrightarrow$ **q**.

- *if and only if* abbreviated **iff.**

TRUTH TABLE FOR

$$p \leftrightarrow q$$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

EXAMPLES:

- “1 + 1 = 3 if and only if earth is flat.”
  TRUE

- “Sky is blue iff 1 = 0”
  FALSE

- “Milk is white iff birds lay eggs.”
  TRUE

- "33 is divisible by 4 if and only if horse has four legs."

  FALSE

- "$x > 5$ iff $x^2 > 25$"

  FALSE

- **Note:**

  Biconditional is logically equivalent to conjunction of two implications:

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

same truth values

REPHRASING BICONDITIONAL:

- $p \leftrightarrow q$ is also expressed as:

  "p is **necessary** and **sufficient** for q"
  "if p **then** q, and **conversely**"
  "p is **equivalent** to q"

Note:

$p \leftrightarrow q$ and $q \leftrightarrow p$ **are EQUIVALENT.**
p iff q and **q iff p** **are EQUIVALENT.**

REPHARASING ENGLISH SENTENCES IN BICONDITIONAL CONECTIVE.

- **If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.**

  You buy an ice cream cone if and only if it is hot outside.

- **For you to win the contest it is necessary and sufficient that you have the only winning ticket.**

  You win the contest if and only if you have the only winning ticket.

- **If you read the news paper every day, you will be informed and conversely.**

  You will be informed if and only if you read the newspaper every day.

- **It rains if it is a weekend day, and it is a weekend day if it rains.**

  It rains if and only if it is a weekend day.

- **This number is divisible by 6 precisely when it is divisible by both 2 and 3.**

  This number is divisible by 6 if and only if it is divisible by both 2 and 3.

TRUTH TABLE FOR $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$

p	q	$p \rightarrow q$	$\sim q$	$\sim p$	$\sim q \rightarrow \sim p$	$(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Note:
The result is **TAUTOLOGY**, It means both the expressions involved in statement form are equivalent.

$(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow q)$

p	q	r	$p \leftrightarrow q$	$r \leftrightarrow q$	$(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow q)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	T	F
F	T	T	F	T	F
F	T	F	F	F	T
F	F	T	T	F	F
F	F	F	T	T	T

HIERARCY OF CONNECTIVES

- Resolve Innermost parenthesis, outmost parenthesis,.
- Negation
- Conjunction and Disjunction
- Implication or Biconditional.

$p \wedge \sim r \leftrightarrow q \vee r$

- In this statement form we have no parenthesis.
- Here $p \wedge \sim r \leftrightarrow q \vee r$ means $(p \wedge (\sim r)) \leftrightarrow (q \vee r)$

p	q	r	~r	$p \wedge \sim r$	$q \vee r$	$p \wedge \sim r \leftrightarrow q \vee r$
T	T	T	F	F	T	F
T	T	F	T	T	T	T
T	F	T	F	F	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	T	F	T	F	T	F
F	F	T	F	F	T	F
F	F	F	T	F	F	T

Show that **~p ↔ q** and **p ↔ ~q** are logically equivalent

p	q	~p	~q	~p ↔ q	p ↔ ~q
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	F	F

same truth values

EXERCISE:

- Show that **~(p ⊕ q)** and **p ↔ q** are logically equivalent

p	q	p ⊕ q	~(p ⊕ q)	p ↔ q
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	T	T

same truth values

Laws of Logic

- **Commutative Law:** $p \leftrightarrow q \equiv q \leftrightarrow p$
- **Implication Laws:** $p \rightarrow q \equiv \sim p \vee q$
  $\equiv \sim(p \wedge \sim q)$
- **Exportation Law:** $(p \wedge q) \rightarrow r \equiv p \rightarrow (q \rightarrow r)$
- **Equivalence:** $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- **Reductio ad absurdum** $p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$

APPLICATION:

- **Rewrite the statement forms without using the symbols $\rightarrow$ or $\leftrightarrow$**

  1. $p \wedge \sim q \rightarrow r$
  2. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

- **Solution:**
- $p \wedge \sim q \rightarrow r \equiv (p \wedge \sim q) \rightarrow r$ order of operations

  $\equiv \sim(p \wedge \sim q) \vee r$ implication law

2. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

$(p \rightarrow r) \leftrightarrow (q \rightarrow r) \equiv (\sim p \vee r) \leftrightarrow (\sim q \vee r)$ implication law

$\equiv [(\sim p \vee r) \rightarrow (\sim q \vee r)] \wedge [(\sim q \vee r) \rightarrow (\sim p \vee r)]$

equivalence of biconditional

$\equiv [\sim(\sim p \vee r) \vee (\sim q \vee r)] \wedge [\sim(\sim q \vee r) \vee (\sim p \vee r)]$

implication law

- Rewrite the statement form **$\sim p \vee q \rightarrow r \vee \sim q$** to a logically equivalent form that uses only **~** and $\wedge$.

- **<u>SOLUTION</u>**

$\sim p \vee q \rightarrow r \vee \sim q$	Given statement form
$\equiv (\sim p \vee q) \rightarrow (r \vee \sim q)$	Order of operations
$\equiv \sim[(\sim p \vee q) \wedge \sim (r \vee \sim q)]$	Implication law $p \rightarrow q \equiv \sim(p \wedge \sim q)$
$\equiv \sim[\sim(p \wedge \sim q) \wedge (\sim r \wedge q)]$	De Morgan's law

Show that $\sim(p \to q) \to p$ is a tautology without using truth tables.

- **<u>SOLUTION</u>**

$\equiv \sim(p \to q) \to p$	Given statement form
$\equiv \sim[\sim(p \wedge \sim q)] \to p$	Implication law $p \to q \equiv \sim(p \wedge \sim q)$
$\equiv (p \wedge \sim q) \to p$	Double negation law
$\equiv \sim(p \wedge \sim q) \vee p$	Implication law $p \to q \equiv \sim p \vee q$
$\equiv (\sim p \vee q) \vee p$	De Morgan's law
$\equiv (q \vee \sim p) \vee p$	Commutative law of
$\equiv q \vee (\sim p \vee p)$	Associative law of
$\equiv q \vee t$	Negation law
$\equiv t$	Universal bound law

EXERCISES:

- Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.
- Simplify $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$
- Show that $[B \wedge (B \rightarrow C)] \rightarrow C$ is a tautology.

EXERCISE:

- Suppose that p and q are statements so that $p \rightarrow q$ is false. Find the truth values of each of the following:

- $\sim p \rightarrow q$
  TRUE

- $p \vee q$
  TRUE

- $q \leftrightarrow p$
  FALSE